

PYTHON-SYLLABUS 2025

Part-1 FRONT END

- **WEB**
- **HTML**
 - Element and attributes
 - Basic Tags
 - Heading
 - Anchor
 - Formatting
 - Div
 - Media
 - Types
 - Attribute
 - List and Types
 - Table
 - Colspan
 - Rowspan
 - Forms
 - Element & attribute
 - Inputs
 - Types
 - Attributes
 - Form validation
- **CSS**
 - Styles
 - Inline, internal and External
 - Selectors: universal, Types, class, ID, attribute, pseudo
 - BG properties : color, image, position, size, repetitions
 - Box Model
 - Border, Margin and Padding
 - Display : inline & block
 - Flex
 - Grid
 - Overflow
 - Float & Z-index

- Bootstrap
 - CDN
 - Button
 - Nav Bar
 - Responsive Design
- Transition and Animation.

Part – 2 Python

- **Features & Installation**
 - Interpreter
 - Memory Management
- **Foundation**
 - Syntax
 - Comment
 - Variables
 - Identifiers
 - Pep 8
 - Keywords
 - IO operation
 - Print
 - Input
 - Datatypes
 - Numerics
 - Integer
 - Float
 - Complex
 - Strings
 - Single
 - Double
 - Triple
 - Indexing
 - Slicing
 - Negative Indexing
 - Algorithms
 - Boolean

- Type Casting
 - Type()
 - Data type constructors
- Operators
 - Assignment
 - Arithmetic
 - Operator Overloading
 - Multiplying
 - Concatenation
 - Comparison
 - Membership
 - Identity
 - Id()
- Flow Control
 - Decisions
 - Code Blocks
 - Indentation
 - Pass
 - If elif else
 - Nested Decisions
 - Iterations
 - For
 - Range
 - String
 - Pattern problem
 - Else
 - Continue
 - Break
 - While
 - Prime number Problem
- **Data Structures**
 - Sequence
 - Iterable & Indexed
 - Tuple
 - Creation
 - Methods
 - Packing & Unpacking

- List
 - Mutability
 - Methods
 - Set
 - Methods
 - Get Count of Unique chars in a string Problem
 - Dictionary
 - Method
 - Nesting
 - Traversing
 - Data Comprehensions
 - List
 - Set
 - Dictionary
- **Functions**
 - Builtin
 - Map
 - Filter
 - Zip
 - Enum
 - Defined
 - Return
 - Arguments and Parameters
 - Default and Non-default
 - Positional and Keyword
 - Arbitrary and Keyword arbitrary
 - Argument syntax rules
 - Namespace and Scope
 - Recursion
- **Advanced**
 - Lambda
 - Closure
 - Decorator
 - Iterator
 - Generator
- **Applied Python**
 - File Handling

- Modules and Libraries
 - OS
 - Random
 - Datetime
 - Re
 - Request
 - pip
- Exception Handling
- **Object Oriented Programming**
 - Class
 - Objects
 - Custom Exceptions
 - Constructors
 - Inheritance
 - Types
 - MRO
 - Encapsulation
 - Access Specifiers
 - Getter and Setter
 - Abstraction
 - Abstract class
 - Abstract Method
 - Polymorphism
 - Method Overriding
 - Method Overloading
 - Dispatch
 - Multiple Dispatch
- **Concurrency and Parallelism**
 - Multithreading
 - Worker Process
 - GIL
 - Threading module
 - Multiprocessing
 - Asynchronous Programming
 - Asyncio library
 - Coroutine
 - Async

- Await

Part – 3 DATABASE

- **RDBMS Foundation**

- Mysql
- Schema
- DDL
 - Creation Command
 - Datatypes
 - Characters
 - Numbers
 - Datetime
 - Constraints
 - Types
 - PK
 - Alter Commands
 - Drop and Truncate
- DML
 - Insert
 - Update
 - Delete
- DQL
 - Data fetching
 - Where
 - Operators
 - Ordering and Limit

- **COMPLEX OPERATIONS**

- Aggregation
 - Group By
 - Having
- Sub-Queries
- Relations
 - Foreign key
 - Joins
- Views
- Stored Procedures

- Defined functions

Part – 4 WEB TECHNOLOGY

- **NETWORKING 101**

- Request and response cycle
 - OSI Model
 - HTTP and HTTPS
 - IP, DNS and Port
 - Status Codes
- Architectures
 - MVC
 - MVT

- **FLASK**

- Routes
- Render Template
 - Templates and Static
- Redirect
- url for
- Jinja
 - Variables
 - Flow control
 - Inheritance
- URL parameters
- Form validation
 - REST API
- Database Connections
 - DB Sqlite
 - Mysql
 - ORM Sqlalchemy

- **CYBER SECURITY**

- Authentication
 - Sessions and Cookies
 - JWT
- Authorization
 - RBAC
- Vulnerabilities

- **APPLIED PROGRAMMING**
 - Git
 - Postman
 - AWS
- **DJANGO**
 - Installation & Setup
 - File Structure
 - Home page
 - Models
 - Migrations
 - Makemigrations
 - Migrate
 - Middlewares
 - Super User
 - Admin Panel
 - ORM
 - Queryset
 - Prefetch related
 - Select related
 - Signals
 - DRF
 - Pagination
 - Serialization
 - Viewsets